

Decoupling and Integration

Queue, topic, stream or bus. Four shapes, and they are not interchangeable.

A queue message is **consumed and gone**.
A stream record **stays**, and everyone can read it.

The four shapes

SHAPE	SERVICE	USE IT WHEN
Queue	Amazon SQS	Work for exactly one consumer, buffered
Topic	Amazon SNS	Push the same message to many subscribers
Stream	Kinesis, MSK	An ordered log many consumers replay
Bus	Amazon EventBridge	Route events by content, with filtering

ON THE EXAM

SNS pushes and forgets. EventBridge routes on content and integrates with SaaS. Both fan out; only one filters richly.

SQS or Kinesis?

AMAZON SQS

Work, done once

- Message deleted after processing
- One consumer per message
- No ordering unless FIFO
- Scales with no shards to manage

KINESIS DATA STREAMS

A log, read many times

- Records persist for the retention period
- Many independent consumers
- Ordered within a partition key
- Replay from any point in time

THE TIEBREAKER

Several systems needing the same events, or replay after a bug, means a stream. A queue cannot give you either.

SQS details that decide questions

- ◆ **Visibility timeout** hides a message while it is processed. Too short means duplicate processing; too long means slow retries.
- ◆ **A dead-letter queue** catches messages that fail repeatedly, so one poison message cannot block a queue forever.
- ◆ **FIFO** gives ordering and exactly-once processing within a message group, at lower throughput than standard.
- ◆ 256 KB message limit, 14-day retention, and long polling to avoid paying for empty receives.

ON THE EXAM

The same message processed twice usually means visibility timeout shorter than processing time. It is the classic SQS bug.

SNS or EventBridge?

AMAZON SNS

Fan-out, fast and simple

- Very high throughput, low latency
- Filtering on message attributes
- SQS, Lambda, HTTP, email, SMS
- You control who publishes

AMAZON EVENTBRIDGE

Routing on content

- Rules match the event body
- AWS service events arrive free
- SaaS partners and custom buses
- Archive and replay events

THE TIEBREAKER

Reacting to an AWS service event, or routing on the event's content, means EventBridge. Raw fan-out speed means SNS.

Streams: Kinesis and the rest

- ◆ **Shards** set capacity: 1 MB/s in, 2 MB/s out each. On-demand mode removes the sizing decision.
- ◆ **Enhanced fan-out** gives each consumer its own 2 MB/s per shard, instead of sharing it.
- ◆ **Amazon Data Firehose** delivers to S3, Redshift or OpenSearch with no code, buffering for at least 60 seconds.
- ◆ Firehose is **delivery, not processing**. If records must arrive in seconds, use Data Streams with a consumer.

ON THE EXAM

Several consumers each needing full throughput is the enhanced fan-out signal. Without it they share 2 MB/s per shard.

When the answer is a lift and shift

- ◆ **Amazon MQ** runs ActiveMQ or RabbitMQ. Choose it when the application already speaks JMS, AMQP, MQTT or STOMP.
- ◆ **Amazon MSK** is managed Apache Kafka, for existing Kafka producers, consumers and Connect ecosystems.
- ◆ Neither is the answer for a new application. SQS and Kinesis are cheaper and need no brokers.
- ◆ The signal is always an existing protocol or ecosystem that cannot be changed.

ON THE EXAM

A named protocol like JMS or AMQP means Amazon MQ. Named Kafka tooling means MSK. Neither appears without that signal.

Step Functions: Standard or Express?

STANDARD WORKFLOWS

Long, durable, auditable

- Up to one year of execution
- Exactly-once execution
- Full history in the console
- Priced per state transition

EXPRESS WORKFLOWS

Short, cheap, very high volume

- Up to five minutes
- At-least-once execution
- History goes to CloudWatch Logs
- Priced per invocation and duration

THE TIEBREAKER

A human approval step, or anything measured in hours, means Standard. High-volume short-lived orchestration means Express.

Patterns worth naming

- ◆ **Fan-out** is SNS to several SQS queues, so each consumer gets its own buffer and its own retry behaviour.
- ◆ **Load levelling** puts a queue in front of a slow component, so a traffic spike becomes a backlog rather than an outage.
- ◆ **EventBridge Pipes** connects one source to one target with filtering and enrichment, replacing glue Lambda functions.
- ◆ **SQS fair queues**, from 2025, stop one noisy tenant starving others in a shared multi-tenant queue.

ON THE EXAM

Fan-out into queues rather than straight to consumers is the pattern the exam rewards. Each consumer then fails independently.

Reading the scenario

THE SCENARIO SAYS

Several systems need the same events

Replay events after fixing a bug

Route on a field inside the event

Existing JMS or AMQP application

A human must approve, days may pass

Spikes overwhelm a downstream system

IT IS POINTING AT

SNS fan-out, or a stream

Kinesis, or EventBridge archive

EventBridge rules

Amazon MQ

Step Functions Standard

SQS for load levelling

ON THE EXAM

Count the consumers first, then ask whether the message must survive being read.
Those two facts decide most of these.

Meridian Freight's booking service publishes a shipment-created event. Four systems must each receive it: billing, customer notification, warehouse allocation and analytics. Each processes at a different speed, and warehouse allocation is occasionally down for maintenance.

No event may be lost while a consumer is unavailable.

What should the solutions architect design?

- ☐ A SNS publishing directly to four Lambda functions
- ☐ B SNS publishing to four SQS queues, one per consumer
- ☐ C One SQS queue that all four systems poll
- ☐ D Four separate publishes from the booking service, one per system

◆ PAUSE AND DECIDE

What should the solutions architect design?

A SNS to four Lambdas

Works while everything is healthy. SNS retries then gives up, so events are lost while warehouse allocation is down.

C One shared queue

A queue message goes to one consumer. Whichever system reads it first, the other three never see it.

B SNS to four SQS queues

Each consumer gets its own buffer and retry behaviour, so a consumer being down means a backlog rather than lost events.

D Four publishes

Couples the booking service to every consumer, so adding a fifth means changing the publisher.

THE SIGNAL

The fan-out pattern is SNS into queues, not SNS into consumers. The queue is what survives a consumer being offline.

A telematics platform ingests 200,000 vehicle position records per second. Three independent teams consume the data: a live map, a fraud model, and an hourly analytics job. After a bug in the fraud model, the team needs to reprocess the last three days of records.

Ordering per vehicle matters.

What should the solutions architect recommend?

- ☐ A SQS FIFO queues, one per consuming team
- ☐ B Kinesis Data Streams with the vehicle ID as partition key, retention set to seven days
- ☐ C SNS fan-out to three SQS queues
- ☐ D Amazon Data Firehose delivering to S3, with each team reading the bucket

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

- A

SQS FIFO per team
FIFO throughput is far below 200,000 per second, and a consumed message cannot be replayed three days later.
- B

Kinesis with 7-day retention
Ordered per partition key, read independently by all three teams, and replayable from any point inside the retention window.
- C

SNS to three queues
Solves fan-out and not replay. Once each queue is drained, those messages are gone.
- D

Firehose to S3
The near miss. S3 gives replay, and Firehose buffers at least 60 seconds, which the live map cannot accept.

THE SIGNAL

Three requirements: many consumers, ordering, and replay. Only a stream provides all three at once.

A security team wants to react automatically when any EC2 instance in the organisation is tagged with Environment=Production but launched without encryption. They also want to receive alerts from a third-party SaaS security product into the same workflow.

The routing must depend on fields inside the event, not just on its type.

What should the solutions architect use?

- ☐ A SNS with message attribute filtering
- ☐ B EventBridge rules on a custom bus, with a SaaS partner event source
- ☐ C CloudTrail delivering to an SQS queue that a Lambda function inspects
- ☐ D Kinesis Data Streams with a Lambda consumer applying the logic

◆ PAUSE AND DECIDE

What should the solutions architect use?

A SNS attribute filtering

Filters on attributes the publisher sets, not on the event body. AWS service events would also need routing into SNS first.

C CloudTrail to SQS and Lambda

Builds the routing engine yourself in code, and adds nothing for the third-party product.

B EventBridge with a partner source

AWS service events arrive natively, rules match fields inside the event body, and SaaS partners publish into the same bus.

D Kinesis with logic in Lambda

Same objection, with a stream added. The filtering logic lives in code that must be maintained.

THE SIGNAL

Three signals: an AWS service event, routing on content, and a SaaS integration. All three are EventBridge and nothing else.

A manufacturer is migrating a plant control system to AWS. It uses ActiveMQ with JMS topics and queues, transactional sessions, and a message selector syntax the vendor supports but will not change. The migration must complete in four months with no application changes. An architect has proposed rewriting it to use SQS and SNS.

What should the solutions architect advise?

- ☐ A Rewrite to SQS and SNS, which are cheaper and fully managed
- ☐ B Amazon MQ running ActiveMQ, so the application connects unchanged
- ☐ C Amazon MSK, since Kafka can emulate JMS semantics
- ☐ D Run ActiveMQ on EC2 instances the team manages

◆ PAUSE AND DECIDE

What should the solutions architect advise?

- A

 Rewrite to SQS and SNS
Cheaper to run and impossible here. The vendor will not change the application, and four months is not a rewrite window.
- B

 Amazon MQ with ActiveMQ
The same broker, managed. JMS, transactional sessions and message selectors all keep working with no code change.
- C

 Amazon MSK
Kafka does not speak JMS. Adapters exist and they change the application, which is the one thing ruled out.
- D

 Self-managed ActiveMQ
Preserves compatibility and hands back patching, failover and broker operations that Amazon MQ provides.

THE SIGNAL

A named protocol the application cannot change is the Amazon MQ signal. It exists almost entirely for this situation.

An insurance claim workflow has eleven steps. Three call external services, one requires a human adjuster to approve, and claims frequently sit waiting for that approval for several days. The business needs to see exactly where any claim is at any moment. Volume is around 4,000 claims a day.

What should the solutions architect recommend?

- ☐ A Step Functions Express workflows, for cost efficiency
- ☐ B Step Functions Standard workflows, with a task token for the approval step
- ☐ C A chain of Lambda functions triggered by SQS queues
- ☐ D EventBridge rules chaining the eleven steps

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

A Express workflows

Express workflows stop at five minutes. A claim waiting days for a human cannot run inside one.

C Lambda chained by queues

Works, and the workflow now exists only implicitly across eleven functions. Nobody can see where a claim is without building that view.

B Standard with a task token

Runs up to a year, so it can wait for the adjuster, and the execution history shows exactly where every claim is.

D EventBridge chaining

Same objection. EventBridge routes events; it does not track the state of a long-running process.

THE SIGNAL

Five minutes is the Express ceiling, and 4,000 a day is low volume, so the cost argument for Express does not apply anyway.

A ticketing site's checkout writes to a legacy inventory system that handles 200 requests per second. During on-sale events, traffic reaches 5,000 per second for about ten minutes. The inventory system currently returns errors and customers see failures.

Replacing the inventory system is a two-year programme already under way.

What should the solutions architect recommend?

- A Auto Scaling on the inventory system to handle the peak
- B An SQS queue between checkout and inventory, with consumers polling at a sustainable rate
- C An ElastiCache layer in front of the inventory system
- D API Gateway throttling, rejecting requests above 200 per second

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

- A** Auto Scaling the legacy system

Legacy systems are usually the ones that cannot scale horizontally, which is why it is being replaced. Nothing suggests it can.
- B** SQS load levelling

The spike becomes a backlog the queue absorbs, drained at 200 per second. Customers get an accepted order rather than an error.
- C** ElastiCache

Caching helps repeated reads. These are inventory writes, which cannot be served from a cache.
- D** Throttle at 200/s

Protects the inventory system by rejecting most customers during precisely the ten minutes that matter commercially.

THE SIGNAL

Load levelling changes what failure looks like. A ten-minute spike becomes a queue that drains, instead of errors customers see.

What to carry out of this module

- ◆ **A queue message is consumed and gone. A stream record stays.** Replay or multiple consumers forces a stream.
- ◆ **Fan-out means SNS into queues,** not SNS into consumers. The queue survives a consumer being offline.
- ◆ EventBridge routes on the event body and speaks to SaaS. SNS is faster fan-out with attribute filtering.
- ◆ **Express workflows stop at five minutes.** A human approval always means Standard.

ON THE EXAM

Amazon MQ and MSK appear only when a protocol or ecosystem cannot change. Without that signal, they are not the answer.